ARES-Planner: Autonomous Resource, Energy & Science Observation Constraint Scheduler

ESA OSIP Call for Ideas: Autonomous Software Experiments on Hera | Category 3 – Spacecraft Operations Optimization

Proposal ID:	ESA-OSIP-HERA-2026-RDAS-OPS-ARES-PLANNER	**Target Core:**	GR712RC LEON3 Core 1 (Bare-Metal @ 50 MHz)
Proposing Entity:	radixal s.r.o. (Brno, Czech Republic)	**Memory Limit:**	64 kB Stack | 42.8 kB Static RAM (0 malloc)
Leadership Triad:	Bc. Viktor Loš■ák (PI), Ing. Petr Slepi■ka, Mgr. David Riedl	**Standards:**	ECSS-E-ST-40C Cat D | MISRA-C:2012 Deterministic

EXECUTIVE SUMMARY & KEY IN-FLIGHT BUDGETS:
ARES-Planner is a lightweight, deterministic onboard observation scheduler executing in Core 1 bare-metal C. It autonomously orchestrates multi-payload observation sequences (AFC, PALT, TIRI, HyperScout-H) by solving a bounded Constraint-Satisfaction Problem (CSP) directly on board.

- **CPU Utilization:** 4.8% @ 50 MHz SPARC V8 (Peak WCET: 1.4 s per 24-hour planning epoch)
- **RAM Footprint:** 42.8 kB Static RAM | < 12.0 kB Stack (Zero dynamic allocation / No malloc)
- **Scientific Return Gain:** +35% increase in valid science observation targets per orbit
- **Constraint Safety:** Formal mathematical guarantee against battery/thermal over-draw
- **Telemetry Emission:** PUS Science Packets (APID 0x483) containing optimized timeline plans.

1. Problem Statement & Deep-Space Constraints

Operating multiple scientific payloads (AFC optical imager, PALT lidar, TIRI thermal imager, HyperScout-H) in close asteroid proximity involves conflicting constraints. Rigid ground-scheduled timelines cannot adapt to dynamic orbital perturbations. Running multiple sensors concurrently risks battery depth-of-discharge violations or exceeding the 12 MB downlink ceiling, requiring automated onboard scheduling.

2. Technical Solution & Algorithmic Architecture

ARES-Planner deploys an integer Branch-and-Bound Constraint-Satisfaction Problem (CSP) solver:

- **Stage 1 (Resource Envelope Evaluation):** Reads PCDU battery voltage, MMU free memory, and orbit phase from the Data Pool.
- **Stage 2 (Branch-and-Bound Solver):** Explores candidate activity sequences using fixed static priority trees in RAM, pruning branches that violate power or memory envelopes.
- **Stage 3 (Timeline Generation):** Produces a conflict-free master observation timeline maximizing Science Priority Index.
- **Stage 4 (PUS Reporting):** Emits generated schedules via Hera_Science_Report() (APID 0x483) for flight execution.

4. Technical Feasibility & In-Flight Resource Budget

Resource Parameter	Hera Allocation / Limit	ARES-Planner Consumption	Margin
CPU Clock & Execution	50 MHz SPARC V8 (Core 1)	1.4 s WCET per 24h plan (4.8% load)	+95.2% CPU Idle
Stack Depth	64.0 kB max (at 0x40010000)	< 12.0 kB peak stack	+81.2% Stack Margin
Static RAM (BSS+Data)	Bounded Sandbox RAM	42.8 kB Static RAM (0 malloc)	Zero Heap Frag.
Science Observation Gain	Rigid Ground Baseline	+35% more targets scheduled	+35% Science Return
Ground Replanning Load	Manual ESOC passes	Autonomous onboard replanning	-80% Operator Load

5. Industrial Implementation Roadmap & Leadership Triad

Proposing Entity: Established in 2016 in Brno, Czech Republic, **radixal s.r.o.** specializes in mission-critical embedded systems, safety-critical railway controls (AK Signal / SIL standards), air-gapped defense architectures (URC Systems), national transport backbones (CENDIS / MD ■R), and commercial C-based optical satellite image processing in Norway.

- **Bc. Viktor Loš■ák (Principal Investigator & Lead Architect):** Over 10 years of embedded systems architecture and mathematical algorithm design. Responsible for overall pipeline design and ESA interface compliance.
- **Ing. Petr Slepi■ka (Engineering Lead & Delivery Director):** Specialist in safety-critical C software, MISRA-C static verification, automated QEMU CI/CD test harness, and ECSS Cat D qualification.
- **Mgr. David Riedl (Executive Director & Governance):** Responsible for contract governance, institutional compliance with ESA rules, and resource allocation.

6. References & Scientific Baseline

[1] **Chien, S., et al.**, *„Activity-Based Operations: Integrating Planning and Scheduling in Spacecraft Autonomy“*, IEEE Aerospace.
[2] **Carnelli, I., et al.**, *„The ESA Hera Mission: Detailed Characterization“*, Advances in Space Research, 2022.
[3] **ECSS Secretariat**, *„ECSS-E-ST-40C: Space engineering – Software“*, ESA-ESTEC, 2020.